Subject: Mathematics | **PERIODIC TEST – 05 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 2 Long Answer HOTS questions (5 marks each).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A two-digit number with distinct digits is formed using the digits $2, 3$, and 5. The probability that the number formed is divisible by 5 is: **[1]**

Options: (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{1}{6}$

2. In a lottery, there are 10 prizes and 25 blanks. The probability of getting a prize is: **[1]**

Options: (a) $\frac{2}{7}$ (b) $\frac{5}{7}$ (c) $\frac{2}{5}$ (d) $\frac{1}{3}$

3. If the probability of winning a game is 0.07, the probability of losing it is: **[1]**

Options: (a) 0.93 (b) 0.03 (c) 0.70 (d) 0.97

4. A target is in the shape of a circle of radius r enclosed inside a square of side $2r$. A dart is thrown at random and lands inside the square. The probability that it lands inside the circle is: **[1]**

Options: (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{1}{\pi}$ (d) $\frac{\pi}{8}$

5. **Assertion (A):** When two dice are thrown together, the probability of getting a sum of 8 is $\frac{5}{36}$.
Reason (R): The pairs giving a sum of 8 are $(2,6), (3,5), (4,4), (5,3), (6,2)$, which are 5 favourable outcomes out of 36. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A number is chosen at random from the numbers $-3, -2, -1, 0, 1, 2, 3$. What is the probability that $x^2 < 4$? **[2]**

7. A letter of the English alphabet is chosen at random. Determine the probability that the chosen letter is: (i) a vowel (ii) a consonant. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A single die is thrown. Find the probability of getting: (i) a multiple of 2 or 3 (ii) a multiple of 2 and 3 (iii) a number neither divisible by 2 nor by 3. **[3]**

9. Harpreet tosses two different coins simultaneously (one of Rs. 1 and the other of Rs. 2). What is the probability that she gets: (i) at least one head? (ii) at most one tail? (iii) no head? **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [10 Marks]

10. A missing helicopter is reported to have crashed somewhere in the rectangular region of length 9 km and breadth 4.5 km. Inside this rectangular region, there is a lake of length 3 km and breadth 2.5 km located at one corner, and the rest of the region is forest. (i) What is the total area of the rectangular search region? (ii) What is the area of the lake? (iii) What is the probability that the helicopter crashed inside the lake? (iv) What is the probability that the helicopter crashed in the forest region outside the lake? (v) If a search team can clear 4.5 km^2 per day, what fraction of the total area can they cover in 3 days? **[5]**

11. In a game of two standard dice: (i) What is the probability that both dice show odd numbers? (ii) What is the probability that at least one die shows a 5? (iii) What is the probability that neither die shows a 6? (iv) What is the probability that the sum of the numbers is divisible by 4? (v) What is the probability that the product of the numbers is greater than 16? **[5]**

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